



# IndianOil

## **JOURNEY RISK MANAGEMENT (JRM) STUDY**

### **Salem Terminal to SRI MUTHU MUNIESWARA TRADERS**

#### **Objective of the JRM Report**

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### **Regulatory Compliance**

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### **Risk Management Strategy**

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

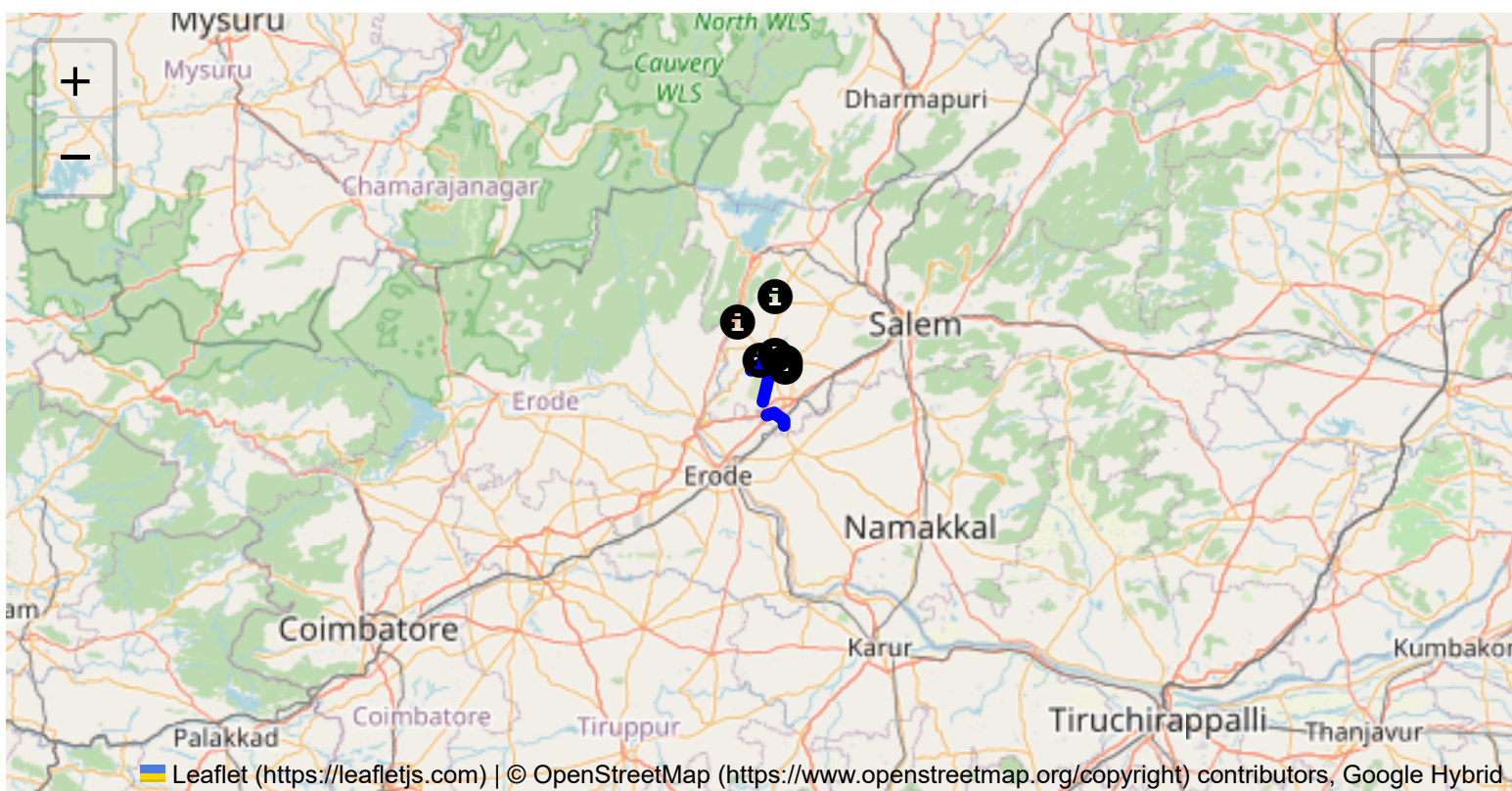
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

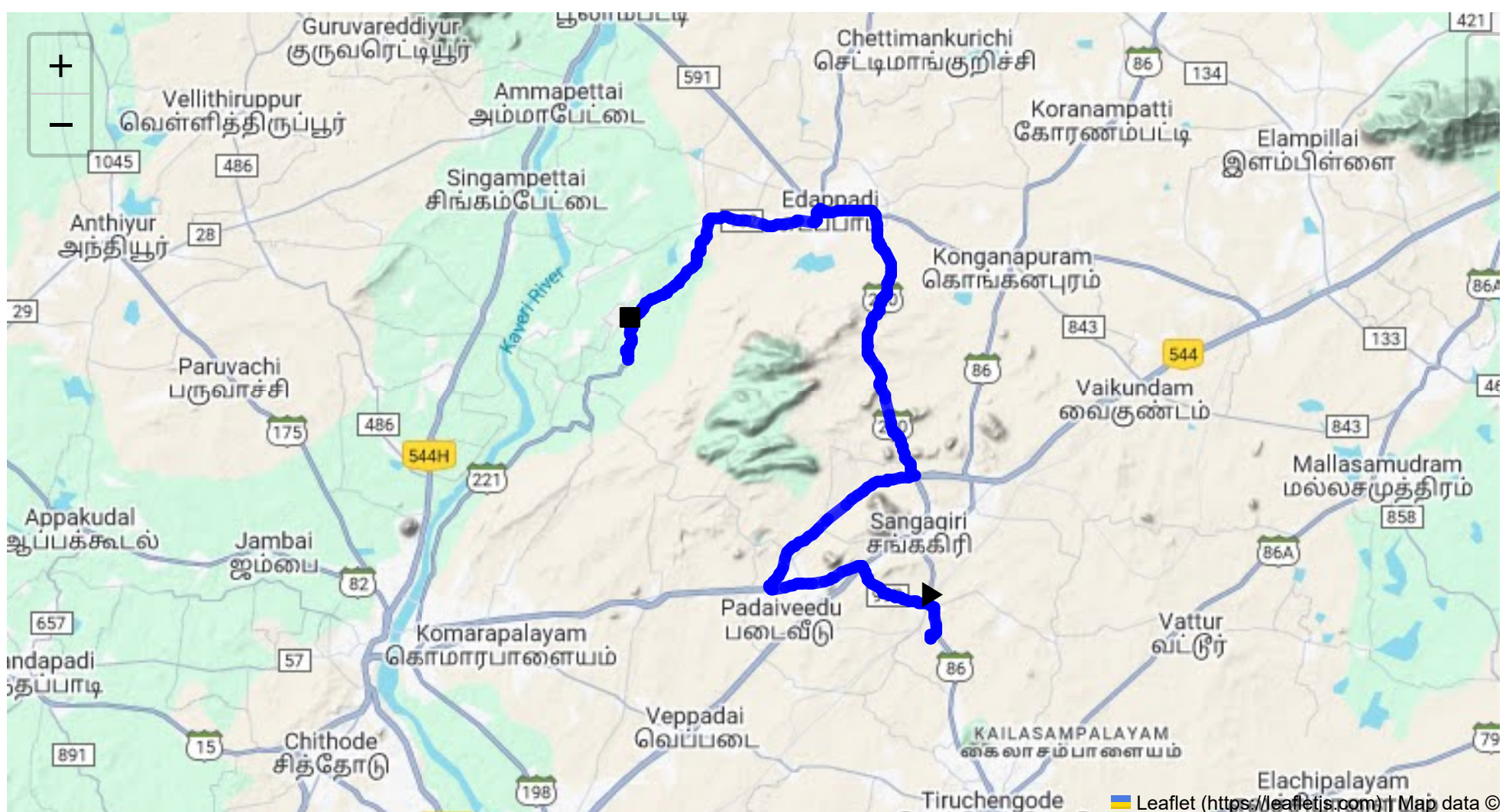
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 40.48 km**  
**Estimated Duration: 1.0 hours**  
**Adjusted Duration (Heavy Vehicle): 1.2 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.533503, 77.767689)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route spans approximately 40.48 kilometers, moving from Sangagiri to Thevur in Tamil Nadu, India. This journey includes major waypoints such as Sankari RS - Erode Privu Rd, Bhavani Main Road (both in Sankari West and Padaiveedu), and Idappadi Rd in Edappadi. These locations suggest a mix of rural roads with some urban intersections, likely featuring state highways and possibly some local roads.

## 2. Typical Weather Conditions

Tamil Nadu enjoys a tropical climate with hot summers, a monsoon season from October to December, and moderate rainfall. Potential weather-related hazards include:

- Heavy monsoon rains which can lead to slippery roads and flooding.
- Summer heat potentially affecting vehicle performance, especially in June to August.

## 3. Traffic Patterns

Traffic density can vary:

- Peak traffic likely occurs during morning (7 AM - 10 AM) and evening (5 PM - 8 PM) hours.
- Congestion-prone areas are expected around urban intersections, particularly in and near Sankari and Edappadi.
- Bhavani Main Road may experience increased congestion due to its role as a key thoroughfare.

## 4. Assessment of Road Quality and Infrastructure

- The main highways like Sankari RS - Erode Privu Rd and Bhavani Main Road are typically maintained but may have variable road quality in some rural sections.
- Infrastructure such as signage and lighting may be lacking in more rural segments.
- Potholes and rough surfaces could be commonplace, especially post-monsoon.

## 5. Suggestions for Alternative Routes

In case of emergencies or adverse conditions, it is advisable to:

- Utilize alternate state highways, if available, that bypass heavily urbanized areas.
- Consider local feedback for the least disrupted alternate routes during severe weather or closures.

## 6. Summary of Local Regulations

- Tamil Nadu adheres to the national regulations concerning the transport of hazardous materials, including restrictions on movement during peak city hours and requirements for vehicle compliance.
- Permits must be secured for transporting hazardous materials, and specific routes might be recommended.

## 7. Historical Incidents Involving Heavy Vehicles

- Historically, some accidents on this route have involved heavy trucks, often due to overloading, speeding, or poor road conditions during monsoons.
- There have been few incidents specific to hazardous materials, but vigilance is advised.

## 8. Environmental Considerations and Sensitive Areas

- The route passes through agricultural zones, requiring careful management to avoid contamination.
- Any spills or emissions must be promptly addressed to prevent damage to local ecosystems.

## 9. Communication Coverage

- Mobile network coverage should generally be good along major roads but may drop in remote, rural areas.
- The use of dual SIMs or wireless communication devices is suggested to mitigate dead zone issues.

## 10. Estimated Emergency Response Times

- Near populated urban areas like Sankari and Edappadi, emergency response times could range from 20-30 minutes.
- Rural segments may experience longer response times, potentially 45 minutes to an hour.

## 11. Overall Summary of Risk Assessment

This route presents moderate risks typical of regional roads in Tamil Nadu, with particular concerns regarding:

- Weather impacts during monsoon.
- Variable road quality affecting heavy vehicles.
- Possible communication outages in rural sections. Mitigation strategies include using GPS and communication tools proactively, adhering to safe driving practices, especially during adverse weather, and considering alternative routes in event of road closures. Transporters of hazardous materials must remain cautious, adhering strictly to regulatory guidelines, and maintain vigilance across sensitive agricultural zones.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.58435, 77.83651 | 15 KM/Hr    |
| 1 | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 2 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 3 | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4 | Turn       | Medium     | 11.45357, 77.85789 | 30 KM/Hr    |
| 5 | Turn       | Medium     | 11.45352, 77.85698 | 30 KM/Hr    |
| 6 | Turn       | Medium     | 11.46048, 77.85036 | 30 KM/Hr    |
| 7 | Turn       | Medium     | 11.46303, 77.84945 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 8  | Turn       | Medium     | 11.46318, 77.84913 | 30 KM/Hr    |
| 9  | Turn       | High       | 11.45542, 77.81725 | 15 KM/Hr    |
| 10 | Turn       | High       | 11.45634, 77.81673 | 15 KM/Hr    |
| 11 | Blind Spot | Blind Spot | 11.49390, 77.86748 | 10 KM/Hr    |
| 12 | Turn       | Medium     | 11.49404, 77.86724 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.50866, 77.86053 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.55039, 77.85511 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.55460, 77.85624 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.58508, 77.85281 | 30 KM/Hr    |
| 17 | Turn       | High       | 11.58532, 77.83454 | 15 KM/Hr    |
| 18 | Turn       | Medium     | 11.58165, 77.83306 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.58089, 77.83231 | 30 KM/Hr    |
| 20 | Turn       | High       | 11.58180, 77.79526 | 15 KM/Hr    |
| 21 | Turn       | Medium     | 11.57111, 77.79256 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.57012, 77.79143 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.55029, 77.77162 | 30 KM/Hr    |
| 24 | Turn       | High       | 11.54528, 77.76829 | 15 KM/Hr    |
| 25 | Turn       | High       | 11.54454, 77.76894 | 15 KM/Hr    |
| 26 | Turn       | Medium     | 11.54350, 77.76802 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.54319, 77.76792 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.53685, 77.76700 | 30 KM/Hr    |

## Emergency Locations

|   | type     | name                             | coordinates            | speed_limit | risk_level |
|---|----------|----------------------------------|------------------------|-------------|------------|
| 1 | hospital | KRS Hospital and Diabetic Centre | 11.5848828, 77.8449664 | 30 km/h     | Medium     |
| 2 | hospital | Kannan Hospital                  | 11.5854802, 77.8425936 | 30 km/h     | Medium     |
| 3 | hospital | Aravind Hospital                 | 11.5854615, 77.8424165 | 30 km/h     | Medium     |
| 4 | hospital | Muthu Eyecare Hospital           | 11.584736, 77.8393794  | 30 km/h     | Medium     |
| 5 | clinic   | Uma Clinic                       | 11.58398, 77.83977     | 30 km/h     | Medium     |
| 6 | clinic   | KGP Clinic                       | 11.5847763, 77.8387488 | 30 km/h     | Medium     |
| 7 | hospital | Government Hospital, Edapadi     | 11.5848264, 77.8380174 | 30 km/h     | Medium     |

|    | type     | name                          | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------|------------------------|-------------|------------|
| 8  | hospital | Sri Sankar Hospital           | 11.5862984, 77.8355021 | 30 km/h     | Medium     |
| 9  | hospital | Sri Advika Hospital           | 11.585049, 77.834422   | 30 km/h     | Medium     |
| 10 | clinic   | Xcellent Homecare Clinic      | 11.584762, 77.8338085  | 30 km/h     | Medium     |
| 11 | clinic   | Raj Physiotherapy Clinic      | 11.5857357, 77.8341218 | 30 km/h     | Medium     |
| 12 | hospital | Dr. O.S. Subramanium Hospital | 11.584996, 77.834235   | 30 km/h     | Medium     |
| 13 | hospital | Aravinthu Eye Hospital        | 11.5866631, 77.8341465 | 30 km/h     | Medium     |
| 14 | hospital | Mohan Hospital                | 11.5841331, 77.8336277 | 30 km/h     | Medium     |
| 15 | hospital | SKM Hospital                  | 11.5841611, 77.8334397 | 30 km/h     | Medium     |
| 16 | hospital | Nivetha Hospital              | 11.5858509, 77.833451  | 30 km/h     | Medium     |
| 17 | clinic   | Krishna Hospital and Clinic   | 11.5846542, 77.8333239 | 30 km/h     | Medium     |
| 18 | hospital | Rathanasamy Hospital          | 11.58583, 77.833443    | 30 km/h     | Medium     |
| 19 | hospital | Pachiappa Hospital            | 11.58571, 77.833131    | 30 km/h     | Medium     |
| 20 | hospital | Arunagiri Hospital            | 11.5837095, 77.8332129 | 30 km/h     | Medium     |
| 21 | clinic   | Rakshan Polyclinic            | 11.5809367, 77.8326223 | 30 km/h     | Medium     |
| 22 | hospital | Siddha Hospital               | 11.5827668, 77.8022539 | 30 km/h     | Medium     |
| 23 | hospital | Government Hospital           | 11.563699, 77.7881936  | 30 km/h     | Medium     |

### Crowded Spots

|   | type   | name                       | coordinates            | speed_limit | risk_level |
|---|--------|----------------------------|------------------------|-------------|------------|
| 0 | school | KRP Matric. Hr. Sec School | 11.4546193, 77.8142445 | 30 km/h     | Medium     |

### Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.58435, 77.83651



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45357, 77.85789



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46048, 77.85036



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46318, 77.84913



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45542, 77.81725



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45634, 77.81673



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.49390, 77.86748



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49404, 77.86724



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.50866, 77.86053



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.55039, 77.85511



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.55460, 77.85624



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.58508, 77.85281



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.58532, 77.83454



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.58165, 77.83306



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.58089, 77.83231



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.58180, 77.79526



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57111, 77.79256



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.57012, 77.79143



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.55029, 77.77162



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.54528, 77.76829



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.54454, 77.76894



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.54350, 77.76802



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.54319, 77.76792



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.53685, 77.76700

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